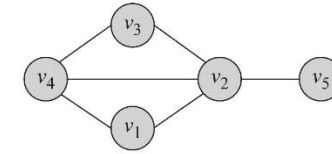


Network Centrality

- **Centrality** in terms of how you connect others
(information broker)

Betweenness Centrality

- How important nodes are in connecting other nodes?
 - Enabling flow of information



- On a social network messages/news being passed from one person to another
- Every pair of nodes exchange a message with equal probability
- Messages take the shortest (geodesic) path
- After a long time, how many messages, on average, would have passed through each vertex en route to their destination?
- The number of shortest path the vertex lies on: **Betweenness centrality**
- Vertices with highest betweenness centrality: (a) the largest number of messages pass, (b) removal from network most disrupt the communication

Betweenness Centrality

Another way of looking at centrality is by considering how important nodes are in connecting other nodes



Linton Freeman

$$C_b(v_i) = \sum_{s \neq t \neq v_i} \frac{\sigma_{st}(v_i)}{\sigma_{st}}$$

σ_{st} The number of shortest paths from vertex s to t – *a.k.a.* **information pathways**

$\sigma_{st}(v_i)$ The number of **shortest paths** from s to t that pass through v_i

Normalizing Betweenness Centrality

- In the best case, node v_i is on all shortest paths from s to t , hence,

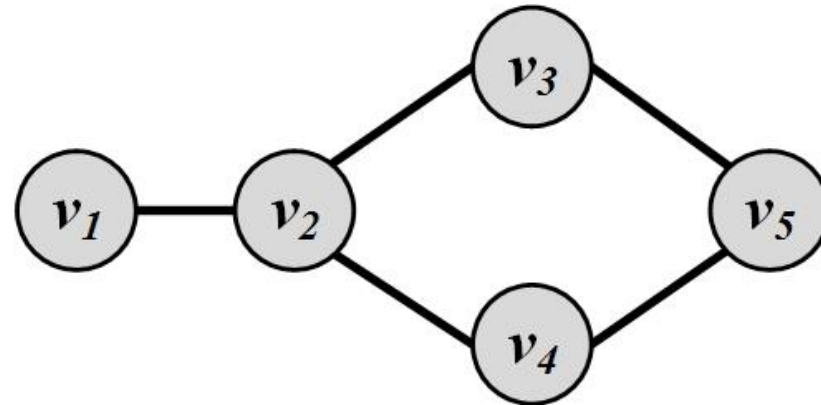
$$\frac{\sigma_{st}(v_i)}{\sigma_{st}} = 1$$

$$\begin{aligned} C_b(v_i) &= \sum_{s \neq t \neq v_i} \frac{\sigma_{st}(v_i)}{\sigma_{st}} \\ &= \sum_{s \neq t \neq v_i} 1 = 2 \binom{n-1}{2} = (n-1)(n-2) \end{aligned}$$

Therefore, the maximum value is $(n-1)(n-2)$

Betweenness centrality: $C_b^{\text{norm}}(v_i) = \frac{C_b(v_i)}{2 \binom{n-1}{2}}$

Betweenness Centrality: Example 1



$$C_b(v_2) = 2 \times \left(\underbrace{(1/1)}_{s=v_1, t=v_3} + \underbrace{(1/1)}_{s=v_1, t=v_4} + \underbrace{(2/2)}_{s=v_1, t=v_5} + \underbrace{(1/2)}_{s=v_3, t=v_4} + \underbrace{0}_{s=v_3, t=v_5} + \underbrace{0}_{s=v_4, t=v_5} \right)$$

$$= 2 \times 3.5 = 7,$$

$$C_b(v_3) = 2 \times \left(\underbrace{0}_{s=v_1, t=v_2} + \underbrace{0}_{s=v_1, t=v_4} + \underbrace{(1/2)}_{s=v_1, t=v_5} + \underbrace{0}_{s=v_2, t=v_4} + \underbrace{(1/2)}_{s=v_2, t=v_5} + \underbrace{0}_{s=v_4, t=v_5} \right)$$

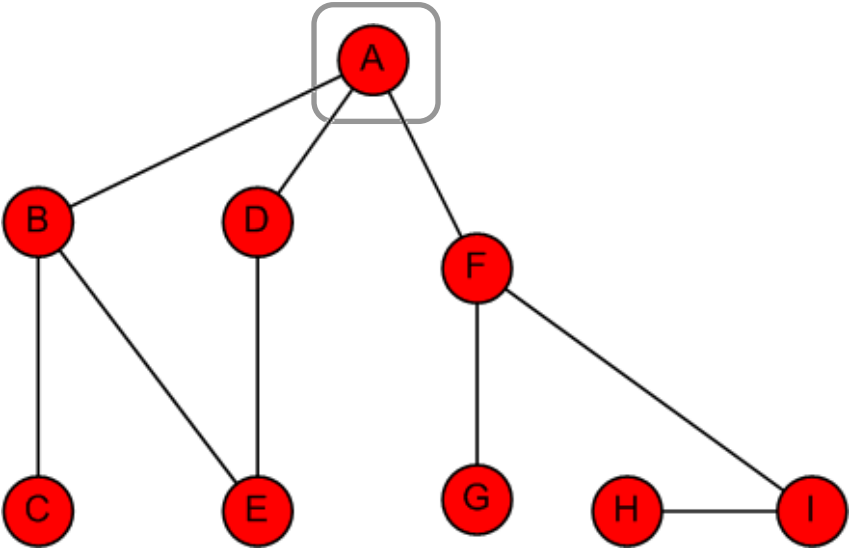
$$= 2 \times 1.0 = 2,$$

$$C_b(v_4) = C_b(v_3) = 2 \times 1.0 = 2,$$

$$C_b(v_5) = 2 \times \left(\underbrace{0}_{s=v_1, t=v_2} + \underbrace{0}_{s=v_1, t=v_3} + \underbrace{0}_{s=v_1, t=v_4} + \underbrace{0}_{s=v_2, t=v_3} + \underbrace{0}_{s=v_2, t=v_4} + \underbrace{(1/2)}_{s=v_3, t=v_4} \right)$$

$$= 2 \times 0.5 = 1,$$

Betweenness Centrality: Example 2



Node	Betweenness Centrality	Rank
A	$16 + 1/2 + 1/2$	1
B	$7 + 5/2$	3
C	0	7
D	$5/2$	5
E	$1/2 + 1/2$	6
F	$15 + 2$	1
G	0	7
H	0	7
I	7	4

Centrality in terms of how fast you can reach others

Closeness Centrality

- The intuition is that influential/central nodes can quickly reach other nodes
- These nodes should have a smaller average shortest path length to others

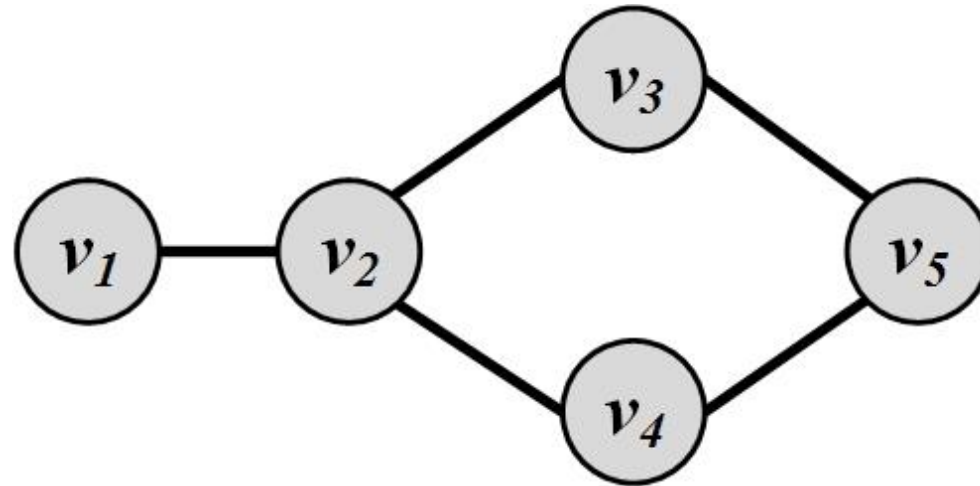


Linton Freeman

Closeness centrality: $C_c(v_i) = \frac{1}{\bar{l}_{v_i}}$

$$\bar{l}_{v_i} = \frac{1}{n-1} \sum_{v_j \neq v_i} l_{i,j}$$

Closeness Centrality: Example 1



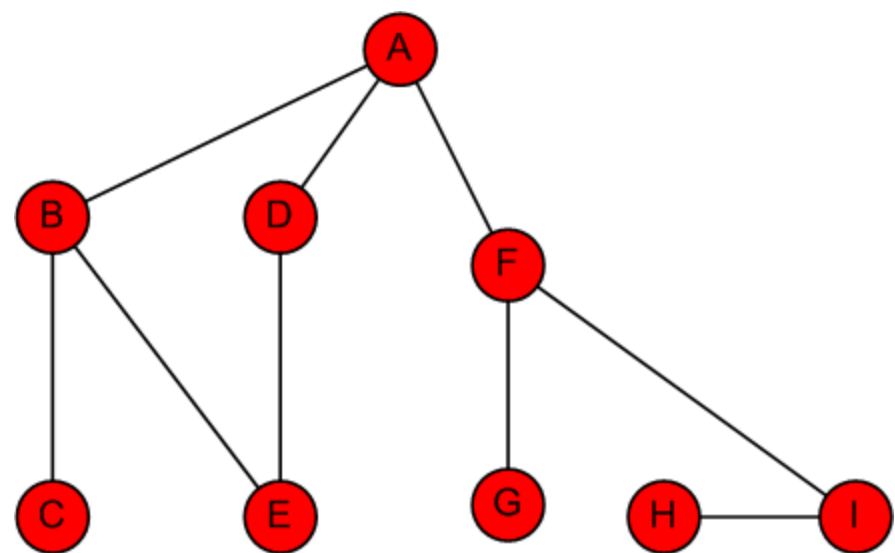
$$C_c(v_1) = 1 / ((1 + 2 + 2 + 3)/4) = 0.5,$$

$$C_c(v_2) = 1 / ((1 + 1 + 1 + 2)/4) = 0.8,$$

$$C_c(v_3) = C_b(v_4) = 1 / ((1 + 1 + 2 + 2)/4) = 0.66,$$

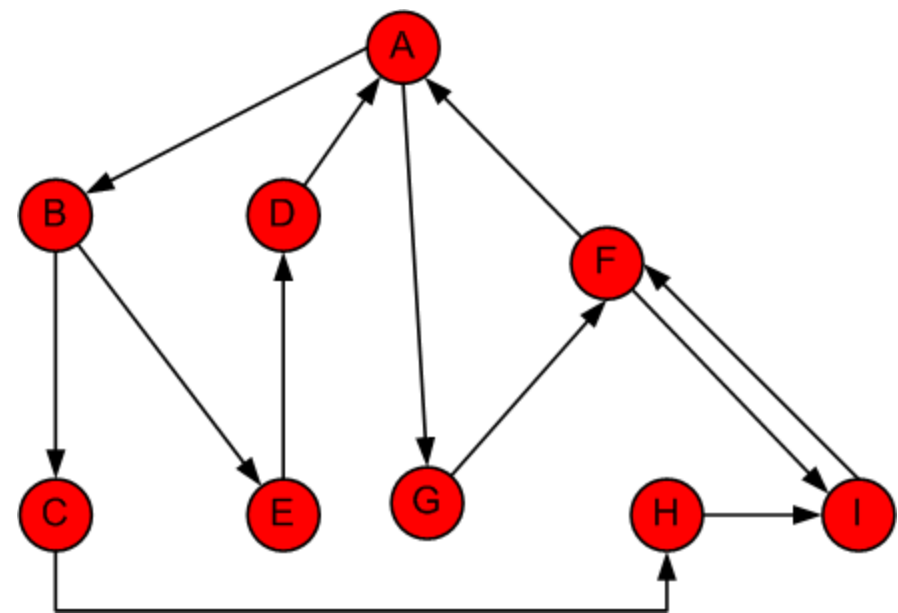
$$C_c(v_5) = 1 / ((1 + 1 + 2 + 3)/4) = 0.57.$$

Closeness Centrality: Example 2 (Undirected)



Node	A	B	C	D	E	F	G	H	I	D_Avg	Closeness Centrality	Rank
A	0	1	2	1	2	1	2	3	2	1.750	0.571	1
B	1	0	1	2	1	2	3	4	3	2.125	0.471	3
C	2	1	0	3	2	3	4	5	4	3.000	0.333	8
D	1	2	3	0	1	2	3	4	3	2.375	0.421	4
E	2	1	2	1	0	3	4	5	4	2.750	0.364	7
F	1	2	3	2	3	0	1	2	1	1.875	0.533	2
G	2	3	4	3	4	1	0	3	2	2.750	0.364	7
H	3	4	5	4	5	2	3	0	1	3.375	0.296	9
I	2	3	4	3	4	1	2	1	0	2.500	0.400	5

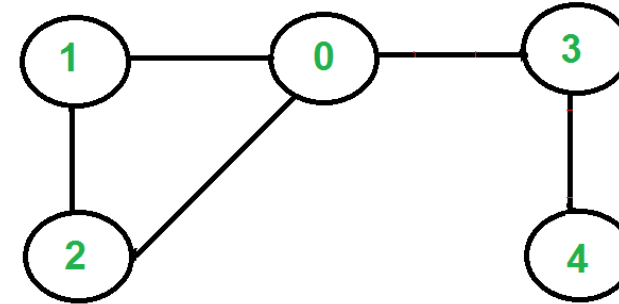
Closeness Centrality: Example 3 (Directed)



Node	A	B	C	D	E	F	G	H	I	D_Avg	Closeness Centrality	Rank
A	0	1	2	3	2	2	1	3	3	2.125	0.471	1
B	3	0	1	2	1	4	4	2	3	2.500	0.400	2
C	4	5	0	7	6	3	5	1	2	4.125	0.242	9
D	1	2	3	0	3	3	2	4	5	2.875	0.348	3
E	2	3	4	1	0	4	3	5	5	3.375	0.296	6
F	1	2	3	4	3	0	2	4	4	2.875	0.348	4
G	2	3	4	5	4	1	0	5	2	3.250	0.308	5
H	4	4	5	6	5	2	4	0	1	3.875	0.258	8
I	2	3	4	5	4	1	4	5	0	3.500	0.286	7

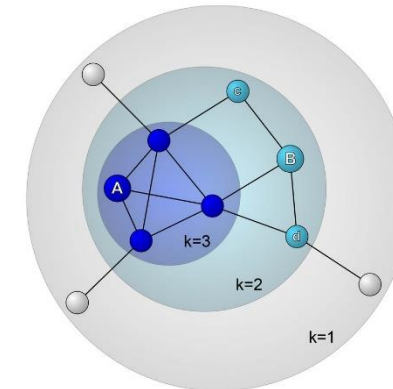
Group centrality: cliques and cores

- A *clique* is a “maximal” subset of the vertices in an undirected network such that every member of the set is connected by an edge to every other.
 - “maximal”: no other vertex in the network that can be added to the subset while preserving the property of connectedness



This Graph can be divided in two Cliques : One Clique contains {0, 1, 2} and other Clique contains {3, 4}

- A *k-core* is a maximal subset of vertices such that each is connected to at least k others in the subset



Friendship Patterns

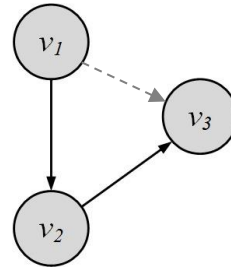
- **Transitivity/Reciprocity**

Transitivity

- Representation:

- For a transitive relation R :

$$aRb \wedge bRc \rightarrow aRc$$



- In a social network:

- v_1 directs a tie to v_2 , and v_2 directs a tie to v_3 , then v_1 also directs a tie to v_3
- ***Transitivity is when a friend (v_1) of my friend (v_2) is my friend***
- Transitivity in a social network leads to a denser graph, which in turn is closer to a complete graph
- We can determine how close graphs are to the complete graph by measuring transitivity

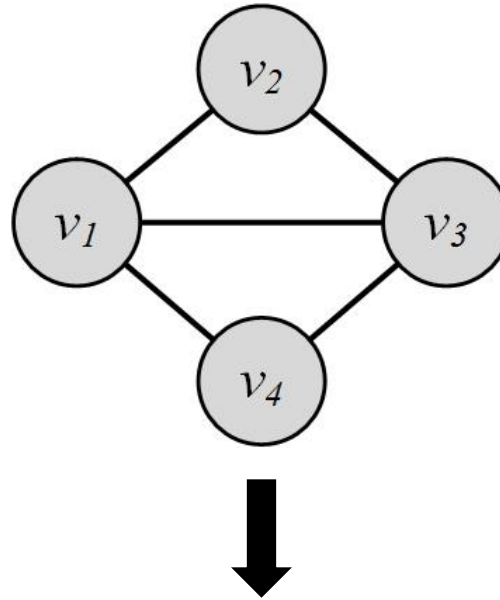
- **Clustering coefficient** measures transitivity in undirected graphs
 - Count paths of length two and check whether the third edge exists

$$C = \frac{|\text{Closed Paths of Length 2}|}{|\text{Paths of Length 2}|}$$

When counting triangles, since every triangle has 6 closed paths of length 2

$$C = \frac{(\text{Number of Triangles}) \times 6}{|\text{Paths of Length 2}|}$$

[Global] Clustering Coefficient: Example



$$C = \frac{\# \text{triangles} \times 6}{\# \text{paths of length two}} = \frac{2 \times 6}{6 + 6 + 2 + 2} = \frac{6}{8} = 0.75$$

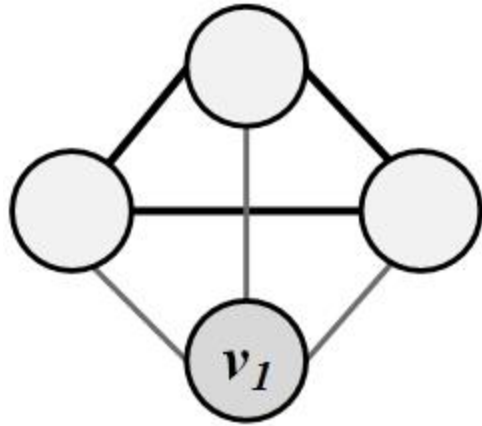
Local Clustering Coefficient

- Local clustering coefficient measures transitivity at the node level
 - Commonly employed for undirected graphs
 - Computes how strongly neighbors of a node v (nodes adjacent to v) are themselves connected

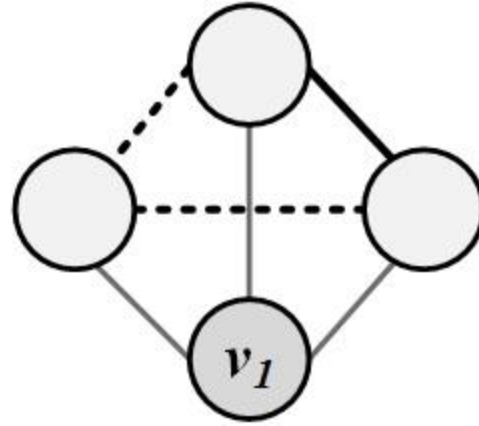
$$C(v_i) = \frac{\text{Number of Pairs of Neighbors of } v_i \text{ That Are Connected}}{\text{Number of Pairs of Neighbors of } v_i}.$$

In an undirected graph, the denominator can be rewritten as: $\binom{d_i}{2} = d_i(d_i - 1)/2$

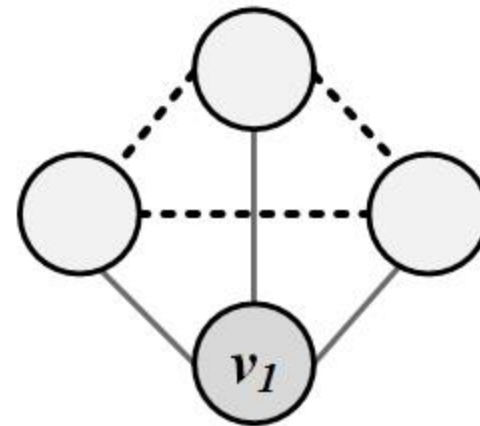
Local Clustering Coefficient: Example



$$C(v_1)=1$$



$$C(v_1)=1/3$$



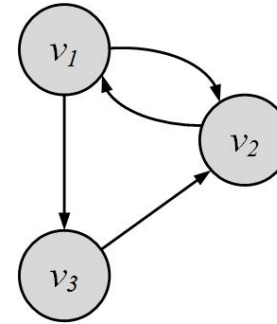
$$C(v_1)=0$$

- Thin lines depict connections to neighbors
- Dashed lines are the missing link among neighbors
- Solid lines indicate connected neighbors
 - When none of neighbors are connected $C = 0$
 - When all neighbors are connected $C = 1$

Reciprocity

*If you become my friend,
I'll be yours*

- Reciprocity is simplified version of transitivity
 - It considers closed loops of length 2
- If node v is connected to node u ,
 - u by connecting to v , exhibits reciprocity

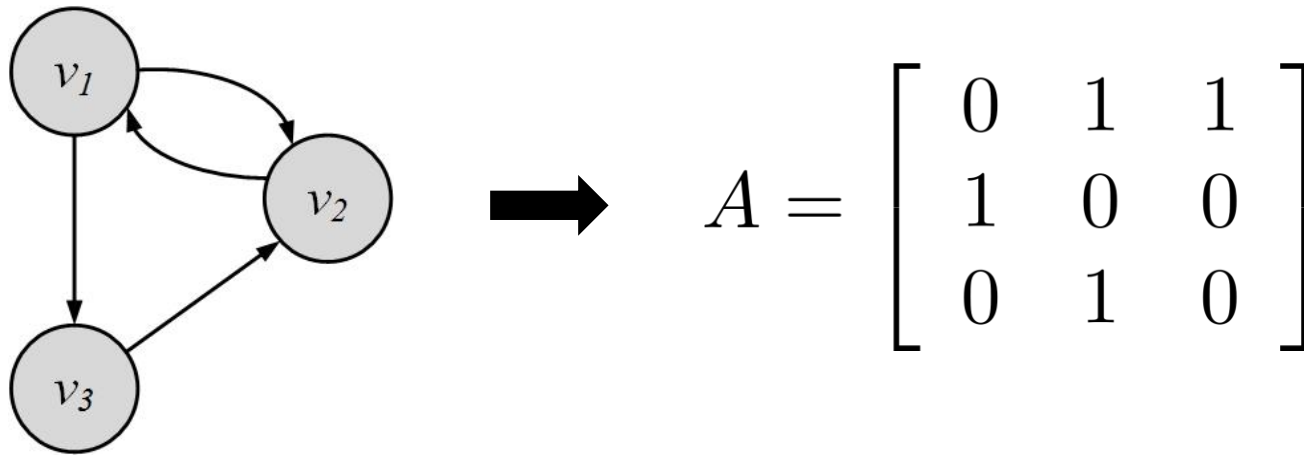


$$\begin{aligned} R &= \frac{\sum_{i,j,i < j} A_{i,j} A_{j,i}}{|E|/2}, \\ &= \frac{2}{|E|} \sum_{i,j,i < j} A_{i,j} A_{j,i}, \\ &= \frac{2}{|E|} \times \frac{1}{2} \text{Tr}(A^2), \\ &= \frac{1}{|E|} \text{Tr}(A^2), \\ &= \frac{1}{m} \text{Tr}(A^2). \end{aligned}$$

What
about
 $i = j$?

$$\text{Tr}(A) = A_{1,1} + A_{2,2} + \cdots + A_{n,n} = \sum_{i=1}^n A_{i,i}$$

Reciprocity: Example



Reciprocal nodes: v_1, v_2

$$R = \frac{1}{m} \text{Tr}(A^2) = \frac{1}{4} \text{Tr} \left(\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \right) = \frac{2}{4} = \frac{1}{2}.$$